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Letter to the Editor

Density–structure correlations in Na_2O – CaO – P_2O_5 – SiO_2 bioactive glasses

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ABSTRACT

Relations for Na_2O – CaO – SiO_2 glasses have been developed to calculate the density of Na_2O – CaO – P_2O_5 – SiO_2 bioactive glasses. The calculation makes use of NMR results of O'Donnell et al. indicating that P_2O_5 forms a separate phase, containing Na_3PO_4 and $\text{Ca}_3(\text{PO}_4)_2$, in the investigated glasses. The volume of the silicate units is the same as that found in Na_2O – CaO – SiO_2 , Na_2O – SiO_2 , and CaO – SiO_2 glasses. Similarly, the volume of PO_4 units is equivalent to that in Na_3PO_4 and $\text{Ca}_3(\text{PO}_4)_2$. Calculated densities are consistent with the experimental data.

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1. Introduction

Correlations between the density and structure has been investigated for various types of silicate glasses such as R_2O – SiO_2 [1,2], RO – SiO_2 [3] and Na_2O – Al_2O_3 – SiO_2 [4,5] (R_2O being an alkali oxide and RO is an alkaline earth oxide). It has been indicated that the volume of structural units in silicate glasses modified with more than one type of modifier oxide is the same as in the corresponding R_2O – SiO_2 or RO – SiO_2 glass. Similar trends were also found in mixed alkali borate glasses [6].

Depending on the kind of glass and the distribution of structural units in the structure, relations were obtained in each case to correlate the glass density with the concentration of structural units and their volume. The predicted relations could be used further to calculate the density of more complicated glasses [7].

In a study on bioactive glasses, Elgayar et al. [8] applied successfully Doweidar model [7] to calculate the density of Na_2O – CaO – P_2O_5 – SiO_2 , K_2O – CaO – P_2O_5 – SiO_2 and Na_2O – K_2O – CaO – P_2O_5 – SiO_2 glasses. The results of NMR indicated that the silicate species in these glasses are Q_3 and Q_2 units (SiO_4 tetrahedra having one and two non-bridging oxygen ions (NBOs), respectively). In all these glasses P_2O_5 is fixed at a relatively low concentration (1.07 mol%). Because the relations in Doweidar model [7] do not account for P_2O_5 and due to that low content of P_2O_5 Elgayar et al. [8] had to ignore the role of P_2O_5 in the glasses they investigated. However they achieved quite good agreement between the experimental and calculated densities.

Recently, O'Donnell et al. [9] studied the density of two Na_2O – CaO – P_2O_5 – SiO_2 bioactive glass series with P_2O_5 content ranged between 0 and 9.25 mol% (Table 1). Series I is characterized with a fixed ratio of $(\text{CaO}/\text{Na}_2\text{O}) = 0.87$. On the other hand, Series II was designed to keep a fixed value for the network connectivity of the glass, assuming formation of isolated orthophosphate units. The content of Na_2O and CaO was adjusted to charge balance the

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Table 1

Composition (mol%) and experimental density (D_m) of the glasses investigated. D_m values are taken from Refs. [9] (Series I and II) and [15] (Series III). Series I is characterized with a fixed ratio of (CaO/Na₂O) = 0.87 and Series II was designed to keep a fixed value for the network connectivity of the glass, assuming formation of isolated orthophosphate units. Series III contains fixed contents of Na₂O and CaO (24.3 and 26.9 mol%, respectively) while the P₂O₅ content increases at the expense of SiO₂. Values of calculated density (D_c) (presented in Fig. 1) are also given.

	SiO ₂ (mol%)	Na ₂ O (mol%)	CaO (mol%)	P ₂ O ₅ (mol%)	D_m (g/cm ³)	D_c (g/cm ³)
Series I [9]						
ICIE1	49.46	26.38	23.08	1.07	2.715	2.701
ICSW2	47.84	26.67	23.33	2.16	2.719	2.700
ICSW3	44.47	27.26	23.85	4.42	2.719	2.700
ICSW5	40.96	27.87	24.39	6.78	2.696	2.699
ICSW4	37.28	28.52	24.95	9.25	2.788	2.698
Series II [9]						
ICSW6	48.98	26.67	23.33	1.02	2.721	2.705
ICSW8	43.66	28.12	24.60	3.62	2.727	2.715
ICSW10	40.71	28.91	25.31	5.07	2.748	2.721
ICSW9	38.14	29.62	25.91	6.33	2.787	2.725
Series III [15]						
H	46.20	24.30	26.90	2.60	2.719	2.729
HP5	43.70	24.40	26.90	5.00	2.721	2.720
HP6.5	42.20	24.30	26.90	6.50	2.687	2.714

phosphate species. Significant differences were observed between the experimental and the calculated densities. The difference increases markedly with increasing the P₂O₅ content. The calculated densities were obtained by using Doweidar model [7] for normalized compositions after excluding the P₂O₅ content. The calculated density obtained by O'Donnell et al. [9] decreases with increasing the P₂O₅ content in Series I whereas it remains without change in Series II. ³¹P MAS-NMR results [10] of the glasses investigated by O'Donnell et al. [9] indicated that P₂O₅ forms a separate phase of Na₃PO₄ and Ca₃(PO₄)₂ in the same proportion of Na₂O and CaO in the glass. On the other hand, ²⁹Si MAS-NMR results revealed that all those glasses contain Q₃ and Q₂ units. These findings were used by O'Donnell et al. [9] to calculate modified densities by taking the effect of P₂O₅ in consideration. Again there were deviations, but at lower extent, between experimental and calculated densities. In the present work the model used for density of Na₂O–CaO–SiO₂ glasses [7] would be extended to be applicable for Na₂O–CaO–P₂O₅–SiO₂ bioactive glasses.

2. Procedure

Instead of separating the orthophosphate components and then normalizing the silicate components, the glass composition would be treated without normalization. In a glass having xNa₂O, yCaO and zSiO₂ (x, y or z being the number of moles of the corresponding oxide in glass) the distribution of species depends on the ratio (x + y)/z. The evidences obtained from NMR investigations on simple silicate glasses [11–14] can be taken as a base to follow the change in concentration of various silicate units.

In the region of $0.5 \leq (x + y)/z \leq 1$ there would be Q₃ and Q₂ units in the glass. Therefore the glass would contain two types of each of such units, depending on the modifier ion. These are Q_{3Na}, Q_{3Ca}, Q_{2Na} and Q_{2Ca} units. As an example Q_{3Na} is a Q₃ unit modified with one Na⁺ ion. The number (n) of units in a glass having xNa₂O, yCaO and zSiO₂ (x, y and z are given as mol, not mol%) can be calculated from the following relations.

$$n_{Q_{3Na}} = 2[xz/(x + y) - x]N_A, \quad (1)$$

$$n_{Q_{2Na}} = [2x - xz/(x + y)]N_A, \quad (2)$$

$$n_{Q_{3Ca}} = 2[yz/(x + y) - y]N_A \quad (3)$$

and

$$n_{Q_{2Ca}} = [2y - yz/(x + y)]N_A, \quad (4)$$

where N_A is Avogadro number.

Although bioactive glasses usually have $0.5 \leq (x + y)/z \leq 1$, it is also adequate to extend the treatment to the region of $1 \leq (x + y)/z \leq 1.5$. The units included in the glass in this region are Q₂ and Q₁, where the latter are SiO₄ tetrahedra having three NBOs. The following relations are predicted for the number of silicate units in such glasses.

$$n_{Q_{2Na}} = [3xz/(x + y) - 2x]N_A, \quad (5)$$

$$n_{Q_{1Na}} = 2[x - xz/(x + y)]N_A, \quad (6)$$

$$n_{Q_{2Ca}} = [3yz/(x + y) - 2y]N_A \quad (7)$$

and

$$n_{Q_{1Ca}} = 2[y - yz/(x + y)]N_A. \quad (8)$$

3. Results and discussion

The glasses studied by O'Donnell et al. [9] (Table 1) would be taken as an example for calculating the density of Na₂O–CaO–P₂O₅–SiO₂ bioactive glasses. In these glasses P₂O₅ forms a separate phase of Na₃PO₄ and Ca₃(PO₄)₂ in the same proportion of Na₂O and CaO in the glass [10]. This means that P₂O₅ has the priority to be fully saturated with modifier oxide, at the expense of SiO₂. To calculate the density it is adequate to start by calculating the amount of Na₂O and CaO consumed to form Na₃PO₄ and Ca₃(PO₄)₂. The amount obtained must be subtracted from the content of Na₂O and CaO in the glass, respectively. The second step is to calculate the amount of Na₃PO₄ and Ca₃(PO₄)₂ in the glass. The rest of Na₂O and CaO can be treated together with SiO₂ using Eqs. (1)–(4) to get the number of Q_n units in the glass. The density (D) of such glasses can be given as

$$D = \sum n_u M_u / \sum n_u V_u, \quad (9)$$

where n_u is number of the structural unit per mole of glass, M_u its mass and V_u its volume.

In addition to the Q_n units the glass includes PO₄ units in Na₃PO₄ and Ca₃(PO₄)₂. As can be noticed there is one PO₄ unit in each Na₃PO₄ molecule, whereas Ca₃(PO₄)₂ has two PO₄ units. In both cases the PO₄ unit has four NBOs, i.e., they are loose units that are not connected to other structural units. The volumes of Q₃ and Q₂ units (Table 2) are given elsewhere [7] and can be used here since the volume of a structural unit in silicate glasses does not depend on the type of glass [1–7]. From the densities of Na₃PO₄ and Ca₃(PO₄)₂ (2.54 and 3.14 g/cm³), respectively [9], the volume of

Table 2Volume of Q_n units and PO_4 units used to calculate the density in the present study. Q_n volumes are taken from Ref. [7].

$V_{Q_{3Na}} (10^{-24} \text{ cm}^3)$	$V_{Q_{2Na}} (10^{-24} \text{ cm}^3)$	$V_{Q_{3Ca}} (10^{-24} \text{ cm}^3)$	$V_{Q_{2Ca}} (10^{-24} \text{ cm}^3)$	$V_{Na_3PO_4} (10^{-24} \text{ cm}^3)$	$V_{Ca_3PO_4} (10^{-24} \text{ cm}^3)$
60.5	79	54.5	66.4	107.2	82

PO_4 units can be obtained as $1.072 \times 10^{-22} \text{ cm}^3$ in Na_3PO_4 and $8.2 \times 10^{-23} \text{ cm}^3$ in $Ca_3(PO_4)_2$. Fig. 1 shows the experimental density values of O'Donnell et al. [9] and the calculated density. The latter has been obtained by using Eq. (9) and the volumes given above.

Fig. 1 shows also the experimental density of Series III [15] and the calculated density. Series III contains fixed contents of Na_2O and CaO (24.3 and 26.9 mol%, respectively) while the P_2O_5 content increases at the expense of SiO_2 . In general, it is shown that the calculated density varies slightly with the glass composition. This is because the variation in composition is relatively limited (Table 1). It can be said that the experimental and calculated densities are consistent, except the end compositions of Series I and II (Table 1). To explain the difference between the experimental and calculated densities of the end compositions of Series I and II O'Donnell et al. [9] assumed the presence of higher density components like amorphous orthophosphate, pyrophosphate or even crystalline phases in the studied glasses. Indeed, XRD patterns [9] indicated that only the glass ICSW4 (Series I, Table 1) has crystalline phases.

The molar volume of a glass is given as

$$V_m = M/D, \quad (10)$$

where M is the molecular relative mass of glass. It can also be correlated to the structural units in glass by the relation

$$V_m = \sum n_u V_u. \quad (11)$$

Fig. 2 shows the experimental V_m values (Eq. (10)) and the calculated values (Eq. (11)) as a function of the P_2O_5 content. With some scattering observed in the experimental data, the agreement is quite good and this reveals the validity of the volumes predicted for PO_4 units in Na_3PO_4 and $Ca_3(PO_4)_2$ phases. Again, in Fig. 2 is shown that the values for the end compositions of Series I and II (Table 1) do not agree with each other.

Fig. 3 presents a comparison between the experimental density [15] of the glasses in Series III (Table 2) and calculated density obtained by different techniques [15–17], in addition to that of the

present work. Lusvardi et al. [15] applied molecular dynamic simulations to derive a descriptor that correlates the change in density with the overall packing degree of structural units in the glass. On the other hand Demkina [16] and Priven [17] used semi-empirical methods based on additive relations, with a specific contribution of each oxide in the glass. These additive relations are however useful to correlate the density with the concentration of oxides in the glass only where the property changes linearly. Within the conventional experimental error limit for the density of such glass types ($\pm 0.02 \text{ g/cm}^3$), Fig. 3 shows an acceptable agreement between the experimental and calculated densities.

The above treatment indicates that the model used for simple silicate- [1–3] and mixed silicate [6,7] glasses can be developed to be applicable for Na_2O – CaO – P_2O_5 – SiO_2 bioactive glasses.

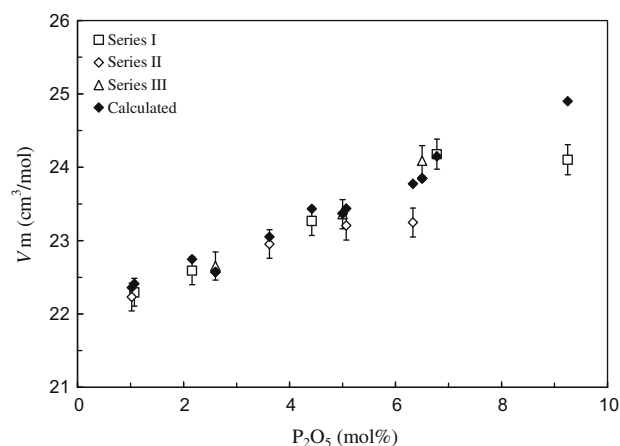


Fig. 2. V_m as a function of the P_2O_5 content in Na_2O – CaO – P_2O_5 – SiO_2 glasses (Table 1). Open symbols represent experimental data ($\pm 0.85\%$) obtained by using Eq. (10) and the density data of O'Donnell et al. [9], Series I ($\square$) and II ($\diamond$), and Lusvardi et al. [15], Series III ($\triangle$). The ($\blacklozenge$)-symbol represents calculated V_m values obtained by using Eq. (11) and the volumes given in Table 2.

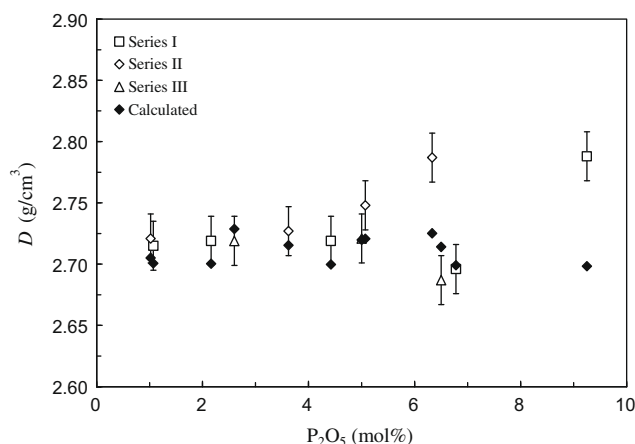


Fig. 1. Density as a function of the P_2O_5 content in Na_2O – CaO – P_2O_5 – SiO_2 glasses (Table 1). Open symbols represent experimental data given by O'Donnell et al. [9], Series I ($\square$) and II ($\diamond$), and Lusvardi et al. [15], Series III ($\triangle$). The ($\blacklozenge$)-symbol represents calculated density obtained by using Eq. (9) and the volumes given in Table 2. Error limit in the experimental densities is taken as $\pm 0.02 \text{ g/cm}^3$.

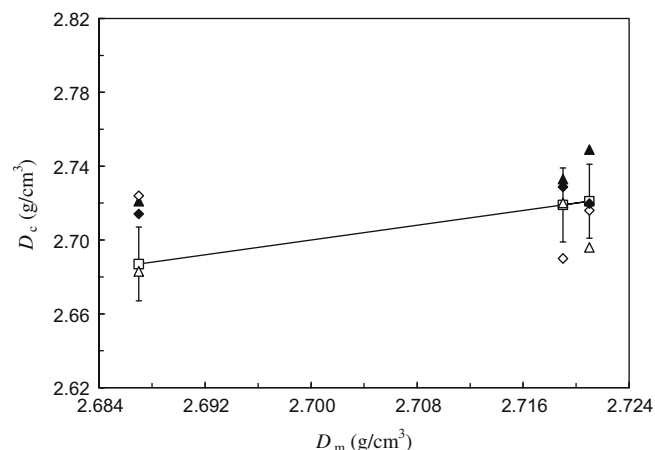


Fig. 3. Correlation between the experimental density (D_m) of Series III (Table 1) and calculated densities (D_c). D_m values ($\square$) are taken from Ref. [15]. D_c values are taken from Refs. [15] ($\diamond$), [16] ($\blacklozenge$), [17] ($\triangle$) and the present work ($\blacklozenge$). The data calculated by Demkina [16] and Priven [17] are taken from Ref. [15].

4. Conclusion

Structural information found by O'Donnell et al. [10] about the incorporation of P_2O_5 in Na_2O – CaO – P_2O_5 – SiO_2 bioactive glasses could be employed to develop relations that can be used for predicting the density of this type of glass. The agreement between experimental and calculated densities of the investigated glass series may be taken as an evidence of the validity of the presented model. Differences found between the experimental and calculated densities of the end compositions of Series I and II (Table 1) might be due to structural modification in the phosphate phase or the cause can be experimental.

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